Zero-Shot, Few-Shot, and Chain-of-Thought Prompting

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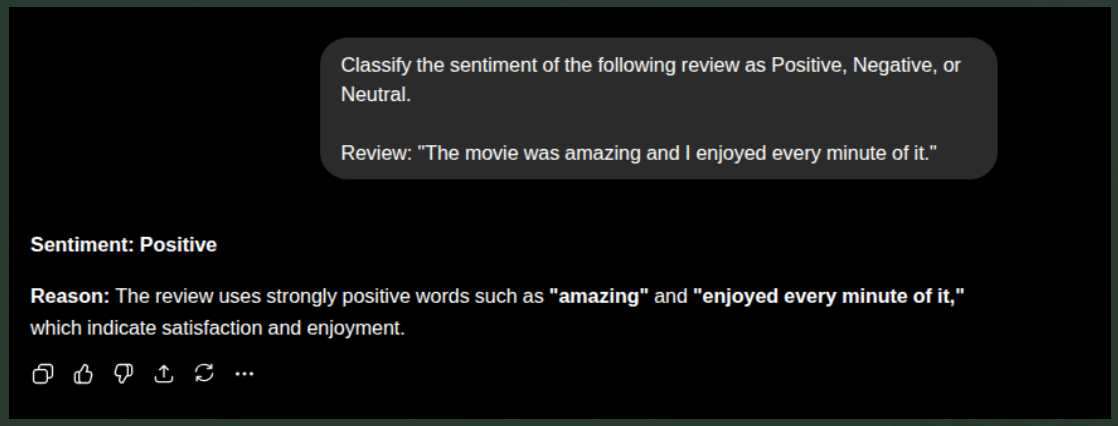
Advanced Prompt Engineering Techniques

Introduction to Prompt Engineering

- • Designing effective prompts for AI models
- • Improves accuracy and relevance
- • Enables complex reasoning and task completion
- • Used in education, coding, research, and business

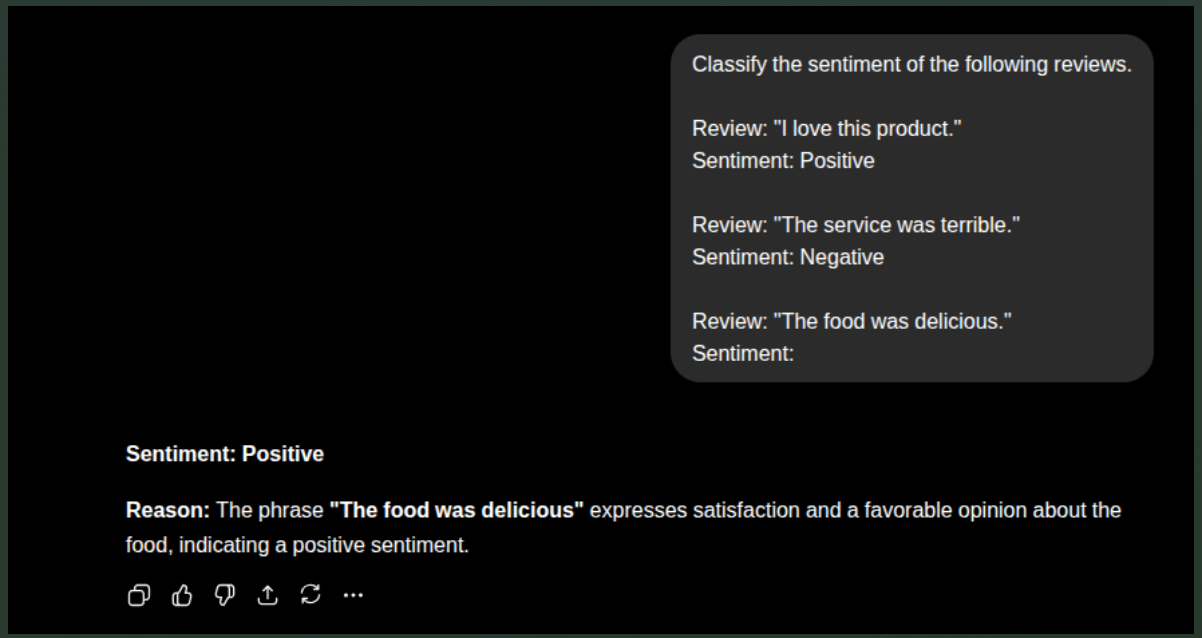
Zero-Shot Prompting

- Definition: Performing a task without examples.
- Example: Classify the sentiment of the following review as Positive, Negative, or Neutral.
- Review: "The movie was amazing and I enjoyed every minute of it."
- Applications: Translation, summarization, Q&A, sentiment analysis.
- Advantages: Simple and fast.
- Limitations: Less reliable for complex tasks.



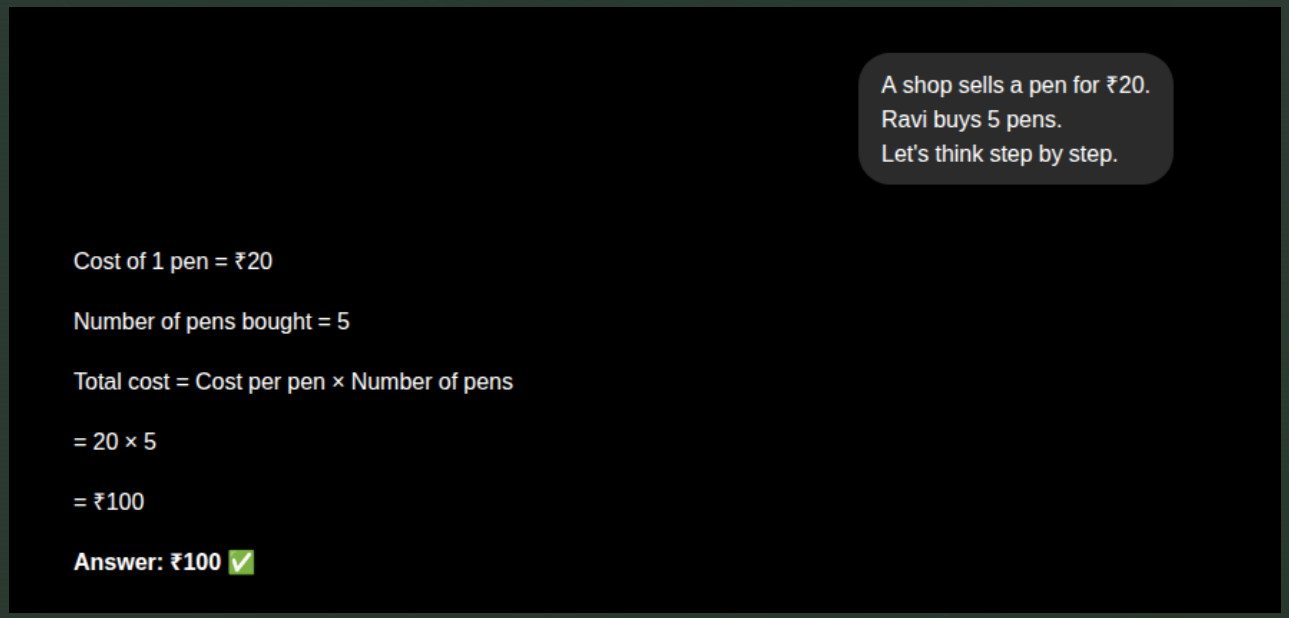
Few-Shot Prompting

- Definition: Providing a few examples before the task.
- Applications: Classification, chatbots, code generation, domain-specific tasks.
- Advantages: Better accuracy and consistency.
- Limitations: Requires carefully chosen examples.



Chain-of-Thought Prompting

- Definition: Encourages step-by-step reasoning.
- Applications: Math, logic, coding, decision-making.
- Advantages: Better reasoning and transparency.
- Limitations: Longer responses and possible reasoning errors.



Comparison of Techniques

- Zero-Shot: No examples, simple tasks.
- Few-Shot: Few examples, structured tasks.
- Chain-of-Thought: Reasoning-focused, complex problems.

Conclusion

- • Zero-shot is simple and fast
- • Few-shot improves performance with examples
- • Chain-of-thought enhances reasoning
- • Choose the technique based on task complexity